

BHUWAN

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Education

National Institute of Technology, Durgapur

Bachelor of Technology in Electrical Engineering

Nov 2022 – Jun 2026

West Bengal, India

Work Experience

Amazon

Jan 2026 – Jun 2026

Software Development Engineer Intern

Bengaluru, India

- Built a **production-scale ML anomaly detection pipeline** for image validation, achieving **>95% accuracy**.
- Designed a **distributed regression testing framework** with pixel-level validation and asynchronous metadata diffing, enabling early defect detection in **pre-deployment CI/CD pipelines**.
- Ran **load tests (1M+ requests)** on ML systems, reducing error rates from **~4% to <0.001%** and improving cost efficiency by **35–50%**.
- Migrated legacy services to a **cloud-native architecture** using scalable, loosely coupled components, improving system throughput, fault tolerance, and horizontal scalability under production load.
- Debugged critical distributed system failures including cross-region authentication latency and clock drifts, using end-to-end tracing to guarantee **strict multi-region data consistency**.

Ciena

May 2025 – Aug 2025

Software Engineer Intern

Gurugram, India

- Built a high-performance **Python automation framework** integrated into Jenkins CI/CD pipelines, optimizing I/O-bound operations to reduce configuration processing time by **40%**.
- Developed a **scalable diff and versioning engine** for configuration artifacts, leveraging algorithmic parsing to boost deployment efficiency and traceability by **30%**.
- Designed and implemented **automated validation pipelines** and CI workflows to verify configuration changes, reducing manual effort by **30%** while improving deployment reliability and consistency.

Amazon ML Summer School

Aug 2025 – Sep 2025

Mentee

Virtual

- Mastered **advanced machine learning workflows**, driving feature engineering and statistical model optimization on large-scale datasets to maximize generalization.

Hacktoberfest

Oct 2025 – Nov 2025

Open Source Contributor

Remote

- Merged **6+ pull requests** across open-source repositories, enhancing runtime performance and refining technical documentation through peer-reviewed code submission

Projects

Smart Issue Routing System | *Next.js, TypeScript, Node.js, Prisma, OpenAI*

Sep 2025

- Built an **LLM-powered issue routing system** with scalable backend services, reducing manual triage by **90%** and reducing response latency.
- Implemented backend services for Role-Based Access Control (RBAC), SLA tracking, and priority handling with real-time dashboards to monitor resolution times and system health.

Refrain Addiction | *React, Node.js, Firebase, MongoDB*

Aug 2024

- Developed a **full-stack platform** with real-time communication and secure workflows, improving user engagement and retention.
- Designed a personalized assessment workflow using behavioral signals and analytics to deliver adaptive recovery insights and track user progress.

Technical Skills

Languages: C++, Java, Python, JavaScript, SQL, HTML, CSS

Core Concepts: Data Structures & Algorithms, System Design, Object-Oriented Programming, Operating Systems, DBMS, Distributed Systems, Computer Networks

Technologies: React, Node.js, Spring Boot, REST APIs, Microservices, MongoDB, Redis

DevOps/Cloud: AWS (EC2, S3, Lambda, DynamoDB), Docker, Kubernetes, Jenkins, Git, Linux

AI/ML: Scikit-learn, TensorFlow, Pandas, NumPy, Machine Learning, Generative AI, LLMs

Achievements

- Competitive Programming:** LeetCode Knight (1928) | Codeforces Expert (1873) | CodeChef 4★ (1975)
- Amazon ML Challenge 2025:** Ranked **64** among **82,000+** participants
- Meta Hacker Cup 2024:** Global Rank **1781**, India Rank **367** (Round 2)